

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 424 — Past Questions & Materials
by Renance

Course: CHM 424
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 400

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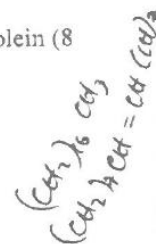
Paper 1 of 3 — CHM 424 - Industrial Chemical Technology

Level 400 • Second Semester • PastQuestion

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DEPARTMENT OF CHEMISTRY
COLLEGE OF SCIENCE
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN, NIGERIA
SECOND SEMESTER EXAMINATIONS 2024/2025
CHM 424: INDUSTRIAL CHEMICAL TECHNOLOGY
INSTRUCTIONS: ANSWER QUESTION ONE AND ANY OTHER THREE QUESTIONS
UNIT:2 CREDITS TIME:2HRS

1. a. What is the difference between petrochemicals and oleochemicals (3 mrks)
 b. Give five examples of raw materials that can be obtained from the biosphere and what they can be used for (10 mrks)
 c. Give the structures and common names of the following fatty acids: (i) C₂₂:6(4,7,10,13,16,19) (ii) 12-hydroxy- 9Z- octadecenoic acid (iii) 5Z,8Z,11Z,14Z - Icosatetraenoic acid (12 mrks)
2. a. A woman making liquid soap goes to the chemical shop to purchase SLS and texapon, use chemical equations to differentiate these two foaming agents and state their names in full (10 mrks).
 b. Give 5 differences between soap and detergents (5 mrks)
3. a. Starch and cellulose are both polysaccharides, use structures to explain their differences (7 mrks)
 b. An organic chemist requires n-hexane to extract some compounds from a plant but he has cassava which was hydrolysed to glucose, use chemical equations to explain how he can obtain n-hexane from the obtained glucose (8 mrks).
4. a. You have just been employed in Guinness Nigeria Plc to produce alcoholic beverages (eg Satzenbrau) and the raw materials available for you are sorghum and barley, what processes are necessary to get the consumable alcoholic drink (10 mrks).
 b. How can sorbitol and gluconic acid be prepared from glucose(use equations) (5 mrks)
5. a. Using cotton plant as a starting material with high cellulose content, how can you obtain a cellulose acetate fabrics used in wedding gowns (use equations) (8 mrks)
 b. Explain the preparation of nanocellulose from coconut husk (7 mrks)
6. a. With a chemical equation, show the hydrogenation of 2-linoleyl-1,3 diolelylglycerol to triolein (8 mrks)
 b. With one example each give the structures of a non-ionic and cationic surfactant (7 mrks)



.....End of Paper.....



Cotton
 rubber
 Petroleum oil
 Cassava



Paper 2 of 3 — CHM424 PQ 2019

Level 400 • Second Semester • PastQuestion

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Paper 3 of 3 — CHM 424 INDUSTRIAL CHEMICAL TECHNOLOGY SECOND SEMESTER 2023/2024

Level 400 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF CHEMISTRY
2023/2024 SESSION, SECOND SEMESTER EXAMINATION

COURSE TITLE: INDUSTRIAL CHEMICAL TECHNOLOGY
COURSE CODE: CHM424 UNIT: 2 TIME: 2 HOURS
INSTRUCTION: ANSWER ANY THREE QUESTION

1a) Differentiate between commodity and specialty chemicals, giving appropriate examples.

b) Discuss the steps and procedures for material balance in chemical process.

c) Explain three (3) different research that can be carried out in a chemical industry.

2a) What are soap and detergent

b) Discuss the chemistry of soap and detergent production.

c) Explain the cleansing action of soap.

✓ 3a) Differentiate between petroleum and petrochemicals.

b) Discuss four (4) petroleum refining processes.

c) Mention six (6) petrochemicals.

✓ 4a) Differentiate between thermoplastics and thermoset polymers, with appropriate examples.

b) Explain two (2) polymerization mechanisms, with appropriate example.

c) Mention four (4) polymer additives and their roles.